University of Bahrain
Collage of Engineering
Department of Chemical Engineering
CHENG490/PICENG490-Senior Project
2021-2022 Semester II

Fuzzy PID (FPID) control for Blood Glucose-Insulin System for T1DM

Group (20)

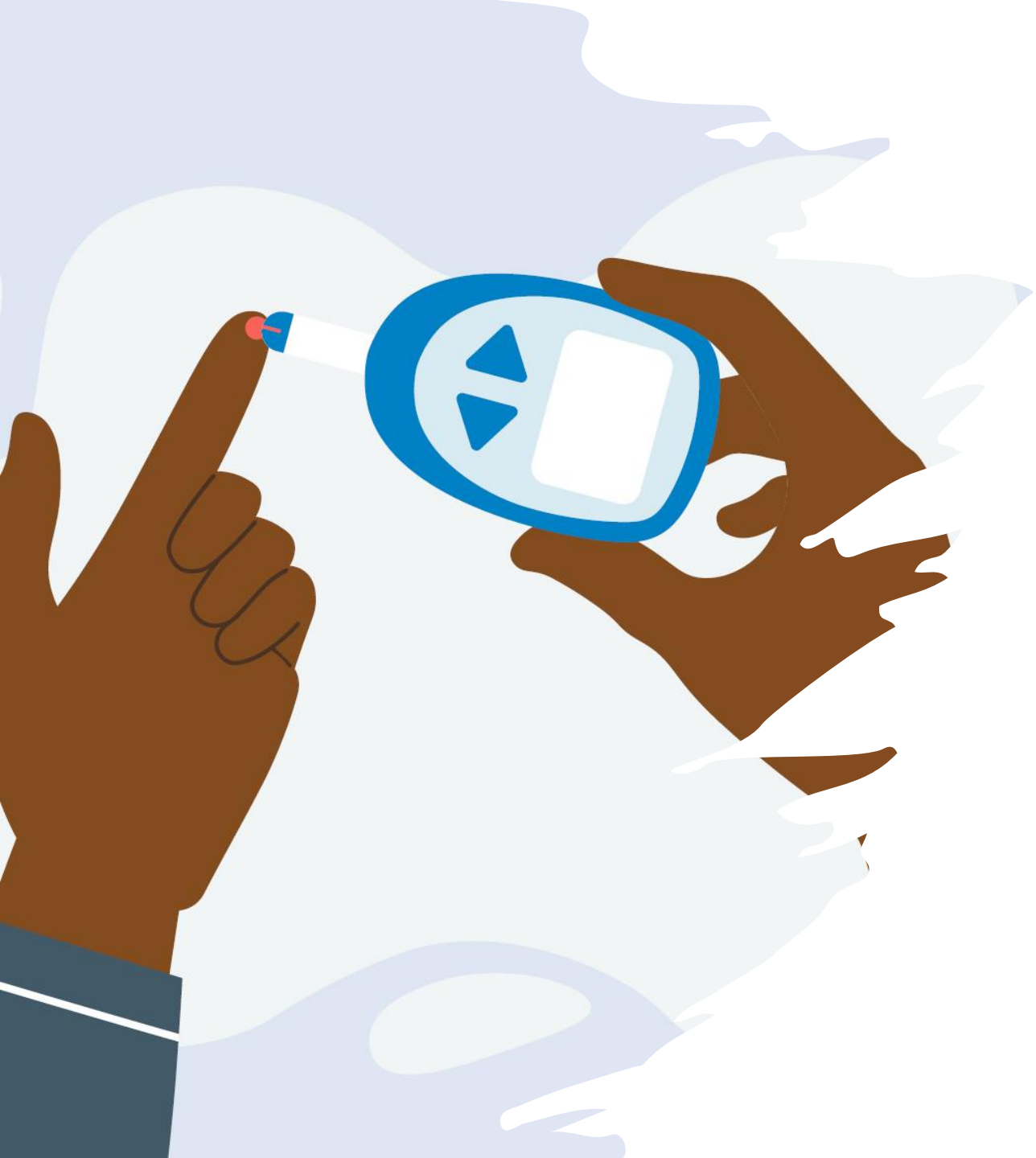
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Outlines

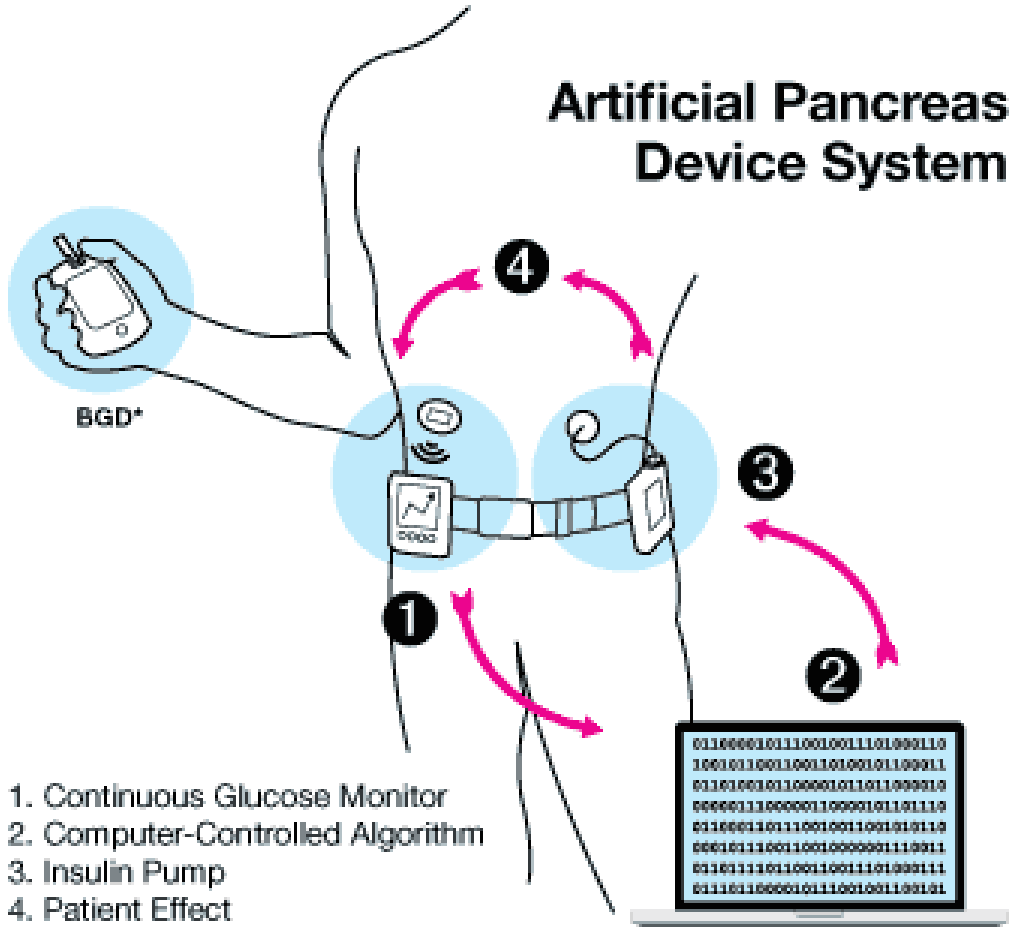
- Introduction
- Bergman Model
- Fuzzy Logic
- Design and Implementation
- Results
- Performance Evaluation
- Conclusion & Future Work

Introduction



Aim of the project is to keep blood glucose levels within an acceptable range by utilizing an external control system, such as an artificial pancreas that uses fuzzy logic.

Artificial pancreas



Bergman Model

Overview

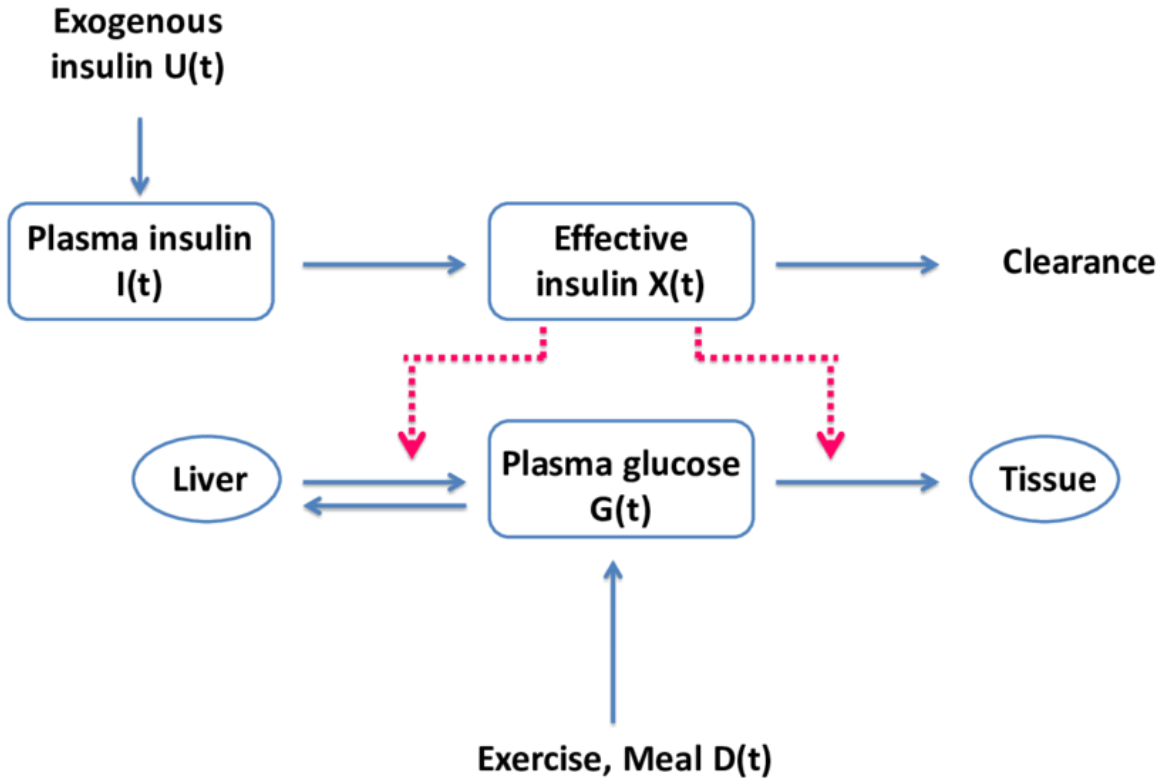
The introduction of the "minimal" model created by Bergman and his colleagues in the late 1970's and early 1980's had a major influence in the field of modeling glucose-insulin dynamics.



Dr. Richard Bergman

What are the model assumptions?

- All required insulin is given externally (U(t)), Thus, the insulin-dependent diabetic patient is modeled.
- From the circulatory system, I(t), a portion of the injected insulin enters the remote compartment, X(t).
- The inaccessible remote insulin (X(t)) actively promotes plasma glucose (G(t)) absorption into the liver and peripheral (extra-hepatic) tissues.



Schematic illustration of Bergman's minimal model

What models capture?

The model consists of three ODEs that capture

I. Plasma glucose dynamics $\frac{dG}{dt}$

II. Plasma insulin dynamics $\frac{dI}{dt}$

III. Insulin concentration in an inaccessible remote compartment $\frac{dX}{dt}$

(which can be conceptualized as interstitial insulin)

Mathematically model given by:

$$\frac{dG}{dt} = -p_1 G - X(G + G_b) + D(t)$$

$$\frac{dX}{dt} = -p_2 X + p_3 I$$

$$\frac{dI}{dt} = -n(I + I_b) + \frac{U(t)}{V_1}$$

Meal

Mathematically given by :

$$D(t) = D \times \exp(a(\log(b \times t) - c)^2)$$

D is the maximum value and **a**, **b**, and **c** are constants that define the slopes and curvature

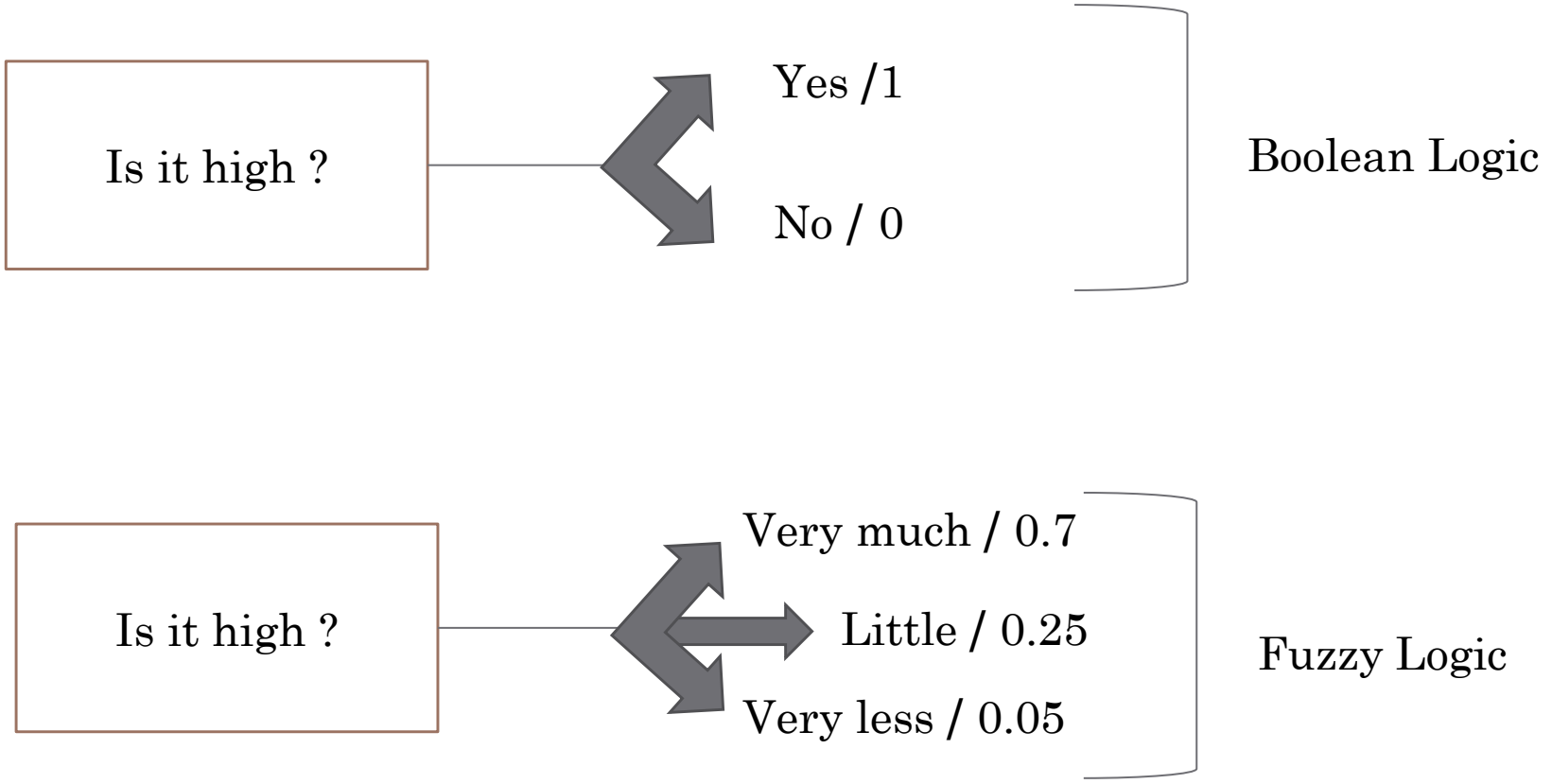
Bergman's Model Parameters Values

Parameter	Meaning	Value
p1	Rate of glucose transport from or to the blood	0.028 (1/min) for normal 0 (1/min) for patient
p2	Rate of insulin transport from the interstitial compartment	0.025 (1/min)
p3	Rate of insulin transport between the blood and the interstitial compartment	$13 * 10^{-6}$
G_b	Basal (180-minute fasting blood glucose level)	4.5 (mmol/L)
I_b	Basal (180-minute fasting blood insulin level)	15 (mU/L)
G_0	Initial glucose level	4.5 (mmol/L)
I_0	Initial insulin level	0 (mU/L)
V_I	Insulin distribution volume	12 L
n	The rate constant represents clearance of plasma insulin	5/54 (1/min)

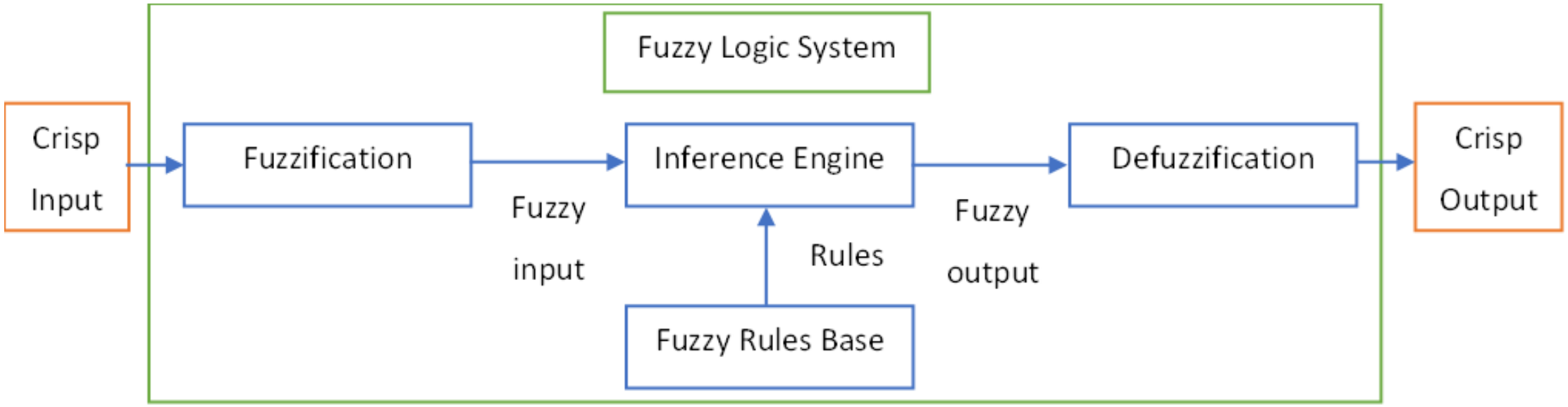
Fuzzy Logic

Fuzzy logic

Glucose :

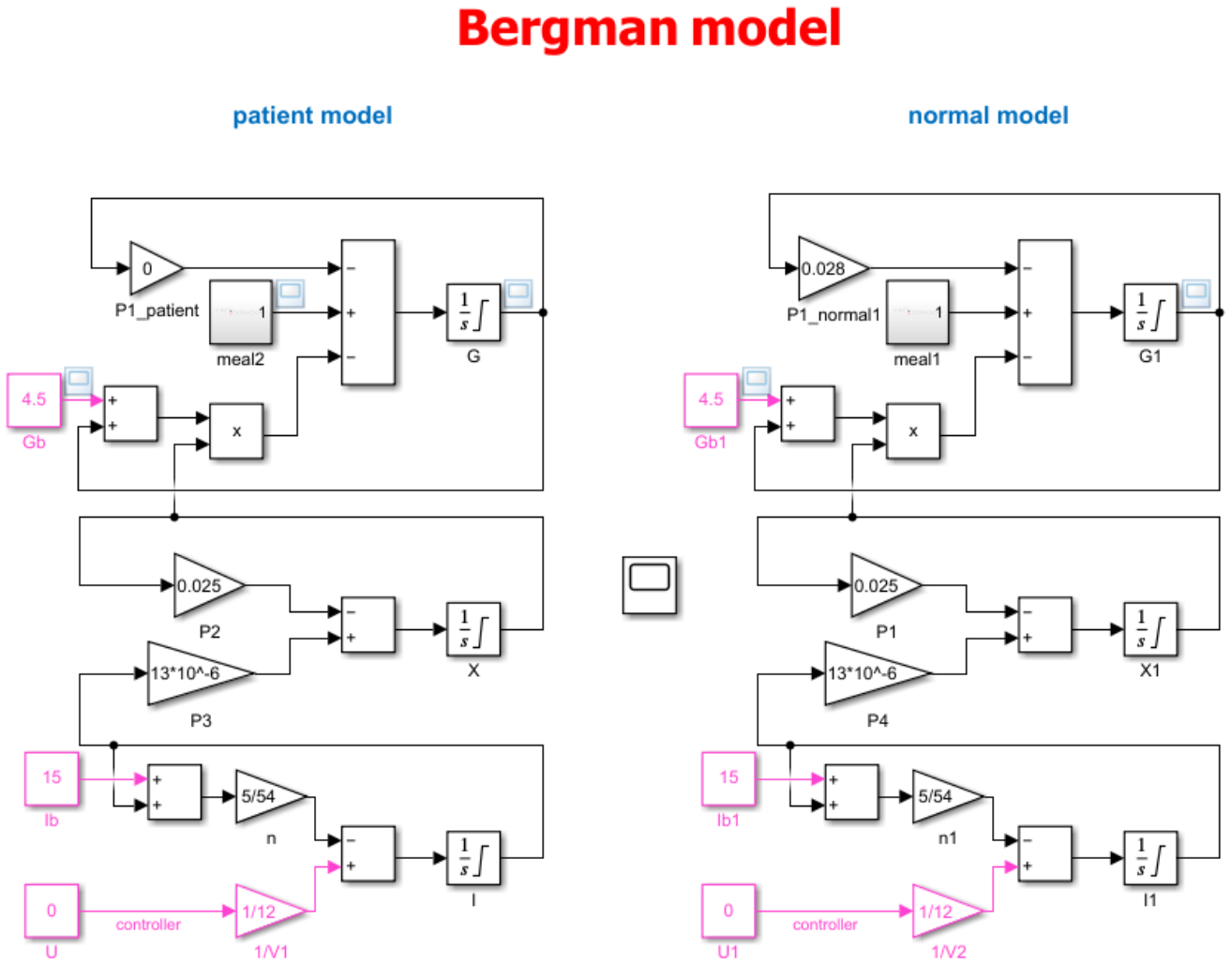


Structure of fuzzy controller

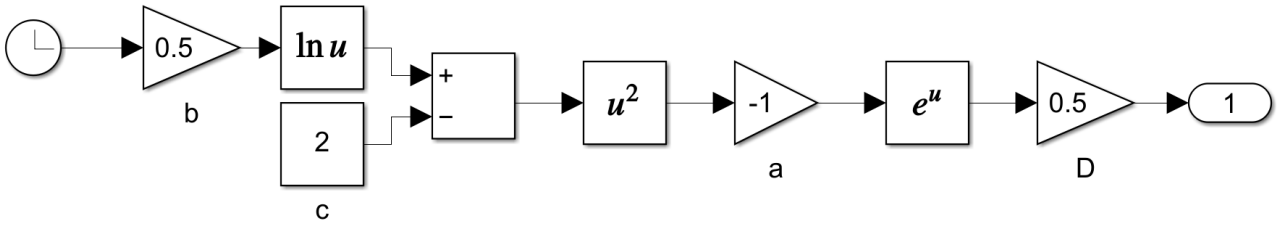


Design and Implementation

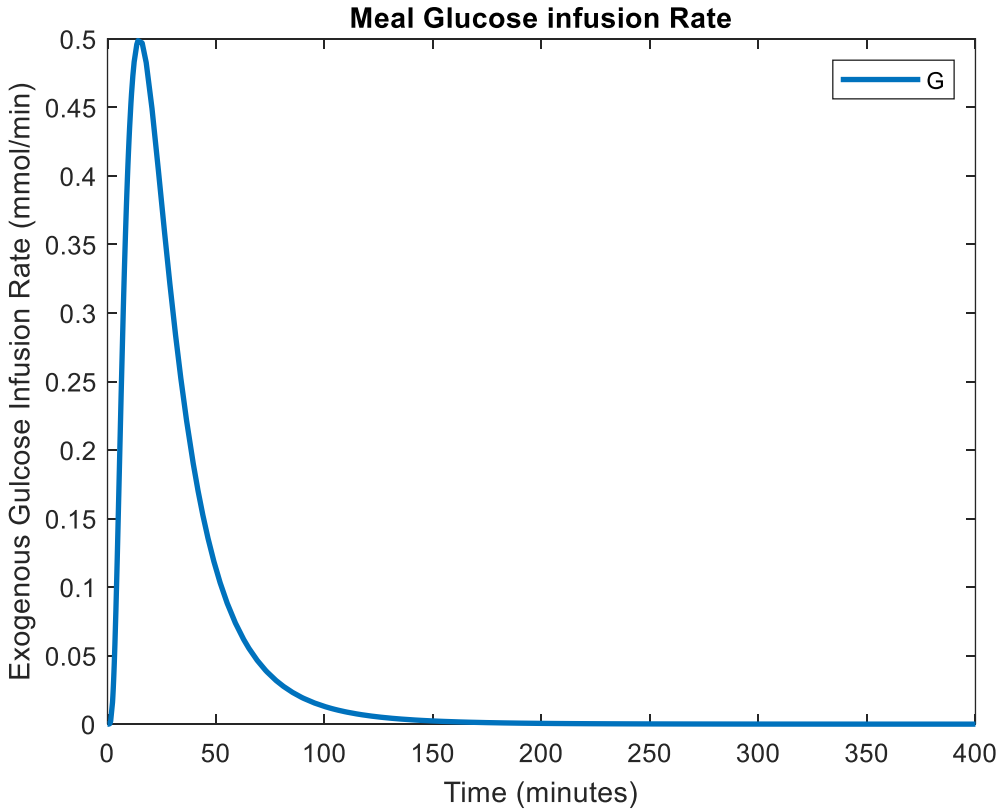
Block diagram of Bergman model for normal and patient



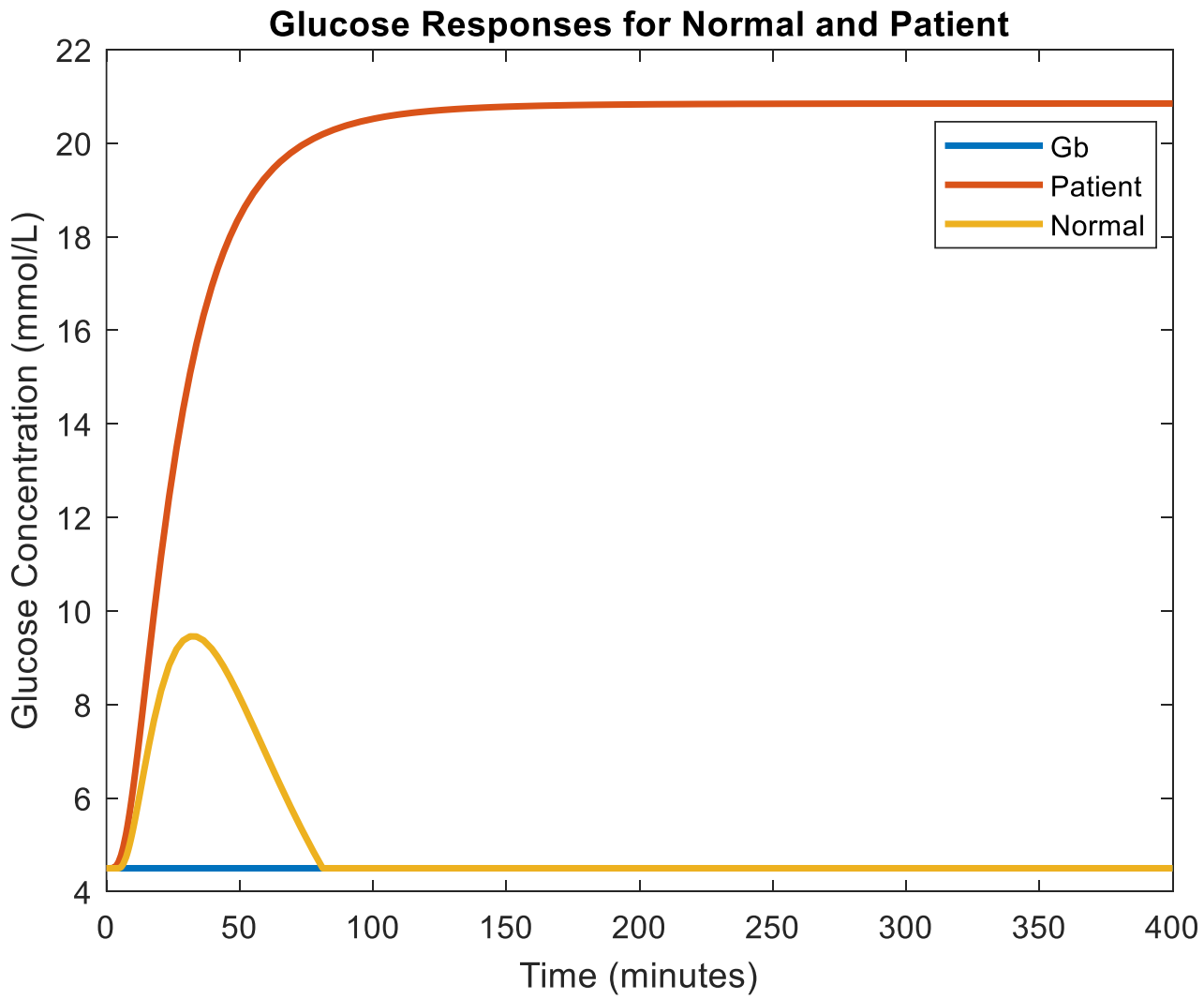
The Meal



The block diagram of the meal

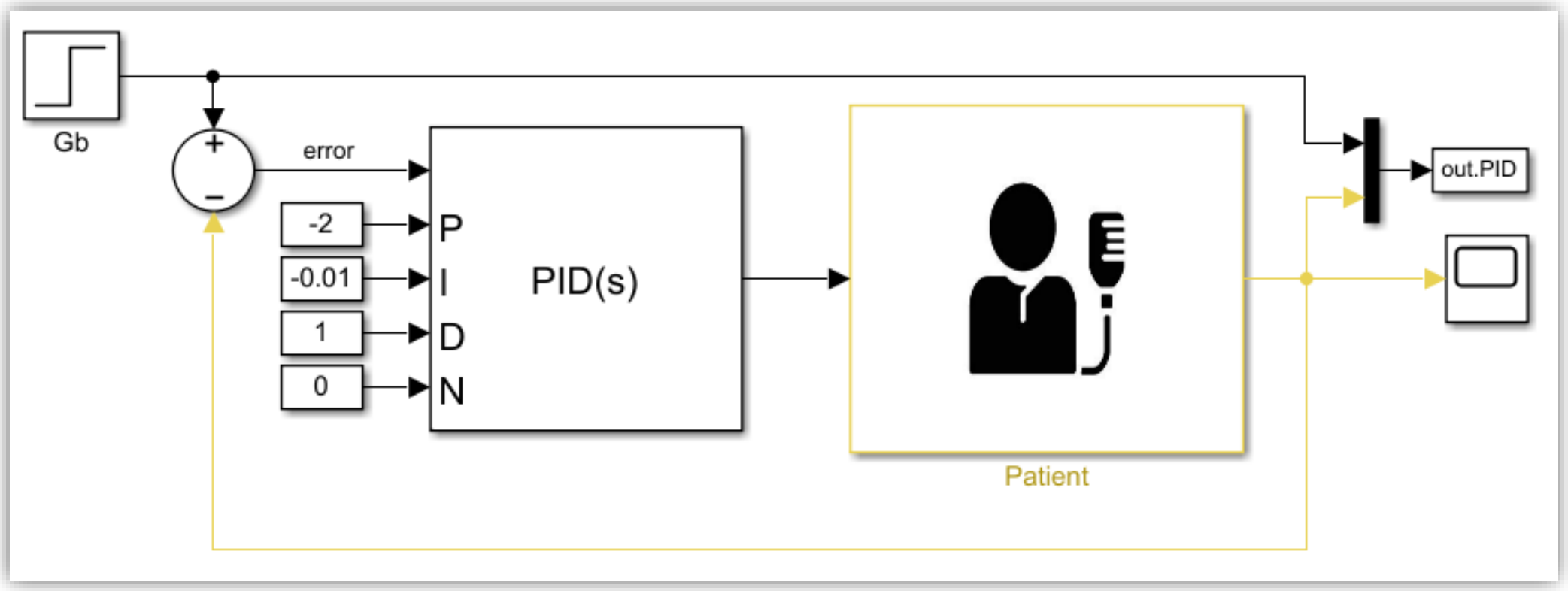


The exponential meal response



Responses for the normal and patient compared with the basal glucose

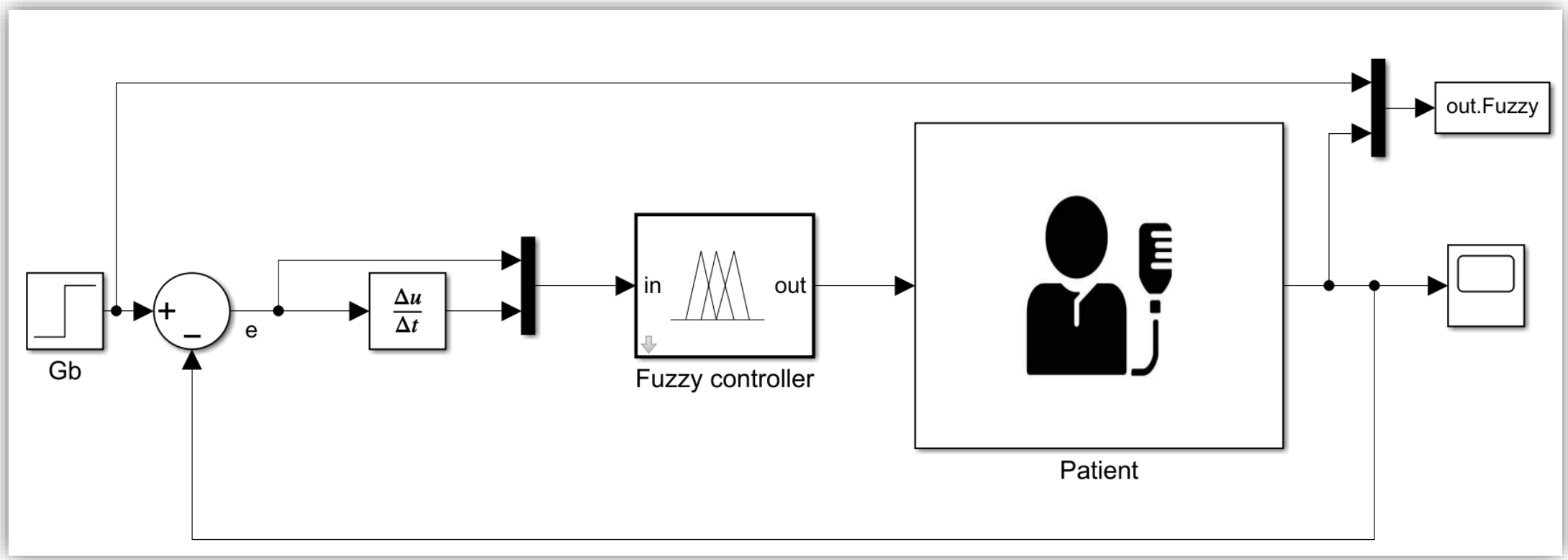
Conventional PID Controller



Block diagram for PID control

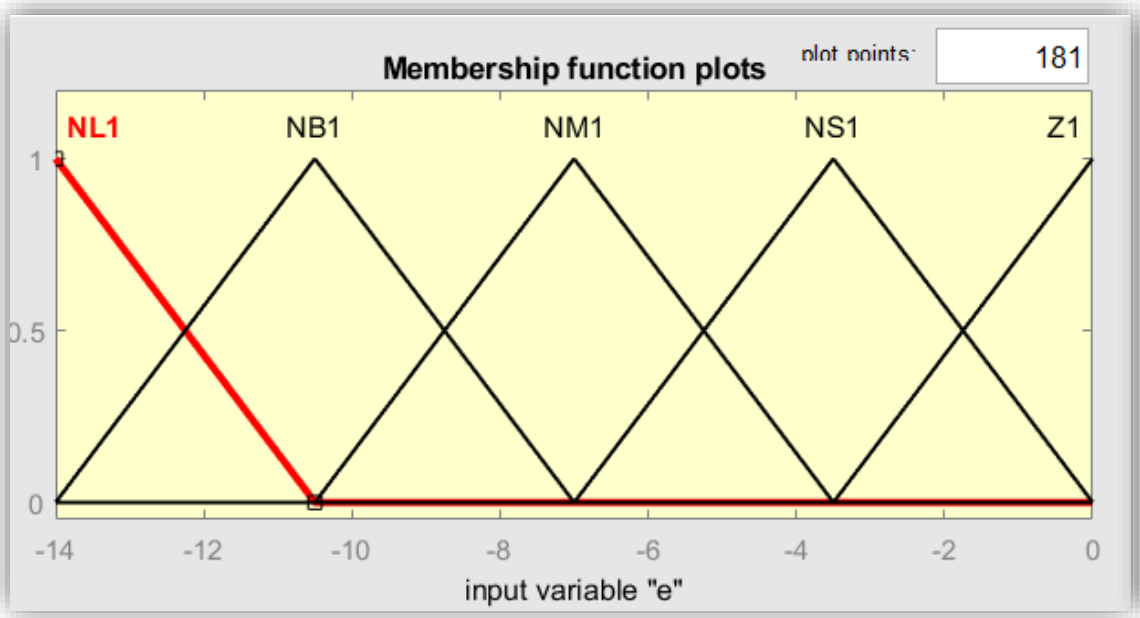
$$u(t) = (-2)\,\mathrm{e(t)} + (-0.01)\int e(t)\,\mathrm{dt} + (1)\frac{de}{dt}$$

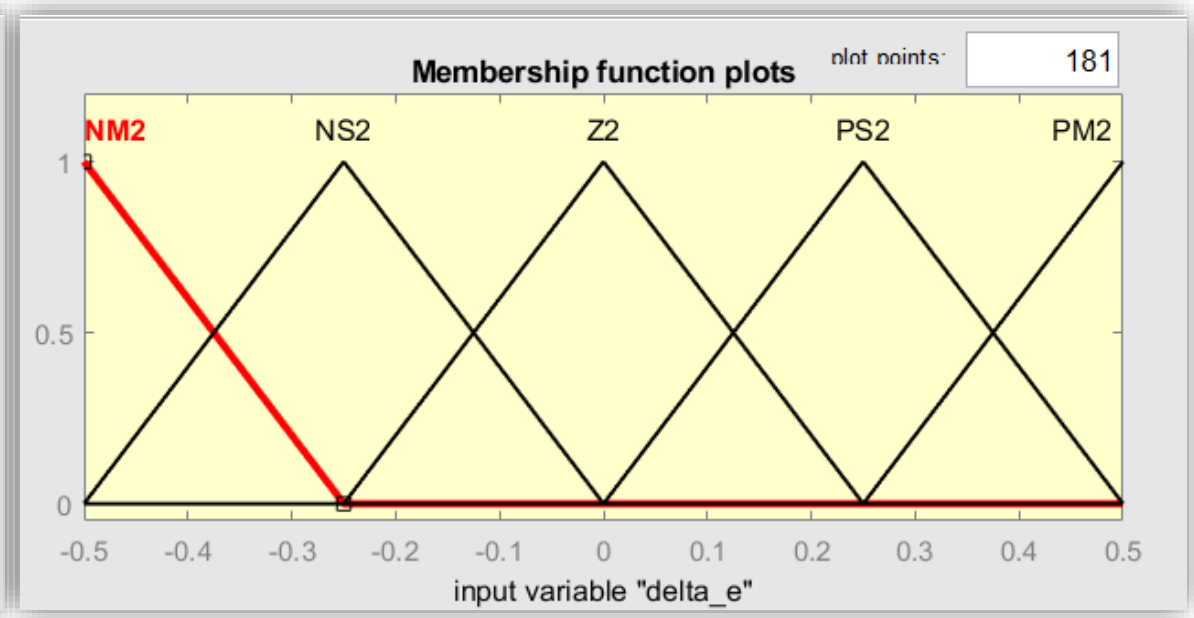
Fuzzy Controller



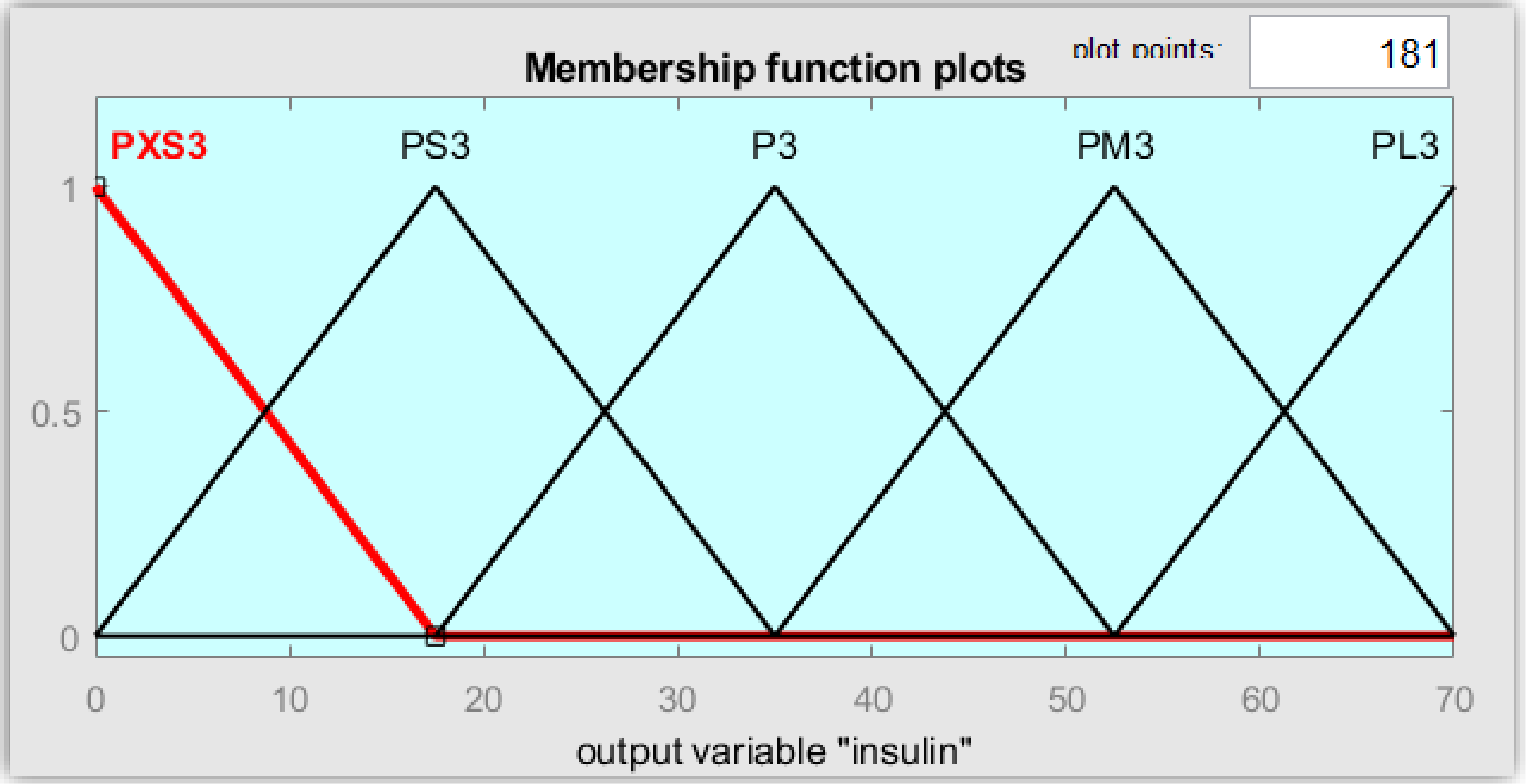
Block diagram of controlling blood glucose using fuzzy controller

Inputs (for fuzzy and fuzzy-PID)





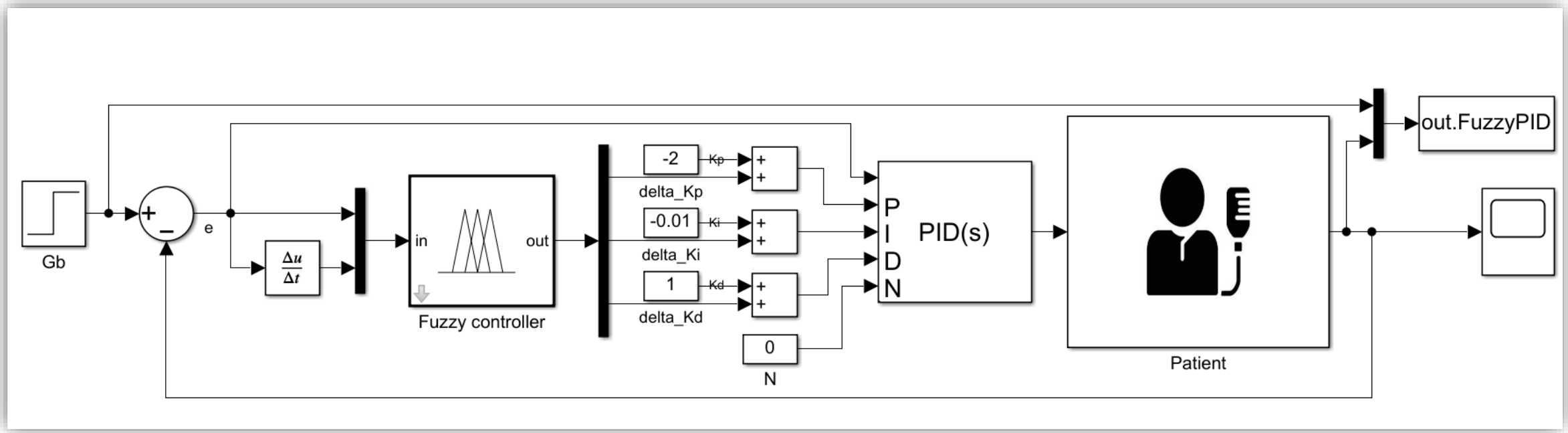
Output (for fuzzy controller)



Rule base for fuzzy controller

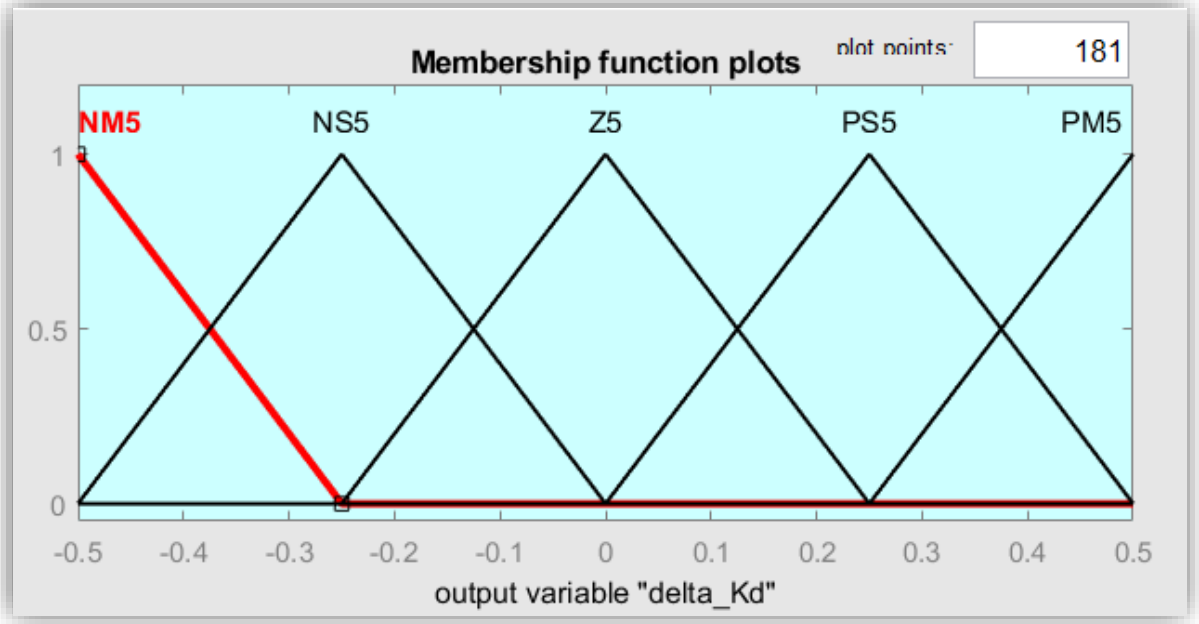
	de					
e		NM2	NS2	Z2	PS2	PM2
	NL1	P3	PS3	PL3	P3	PS3
	NB1	PM3	P3	PM3	PS3	PS3
	NM1	P3	PS3	P3	PXS3	PS3
	NS1	PXS3	P3	PS3	PS3	PXS3
	Z1	PS3	PXS3	PXS3	PXS3	PXS3

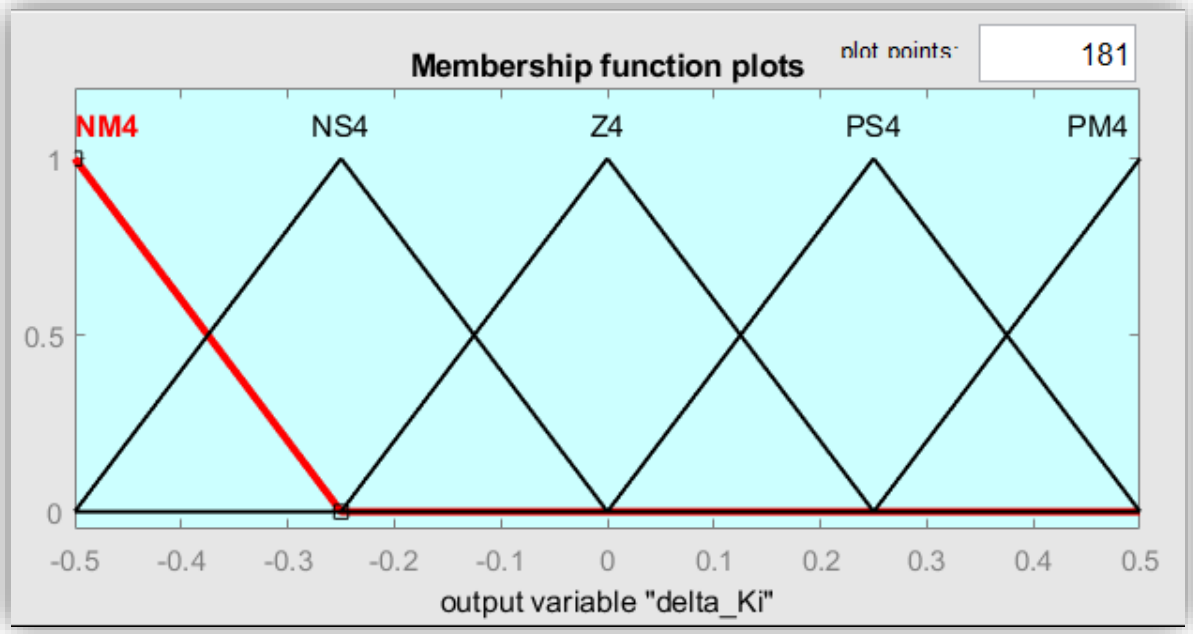
Fuzzy-PID for controlling blood glucose

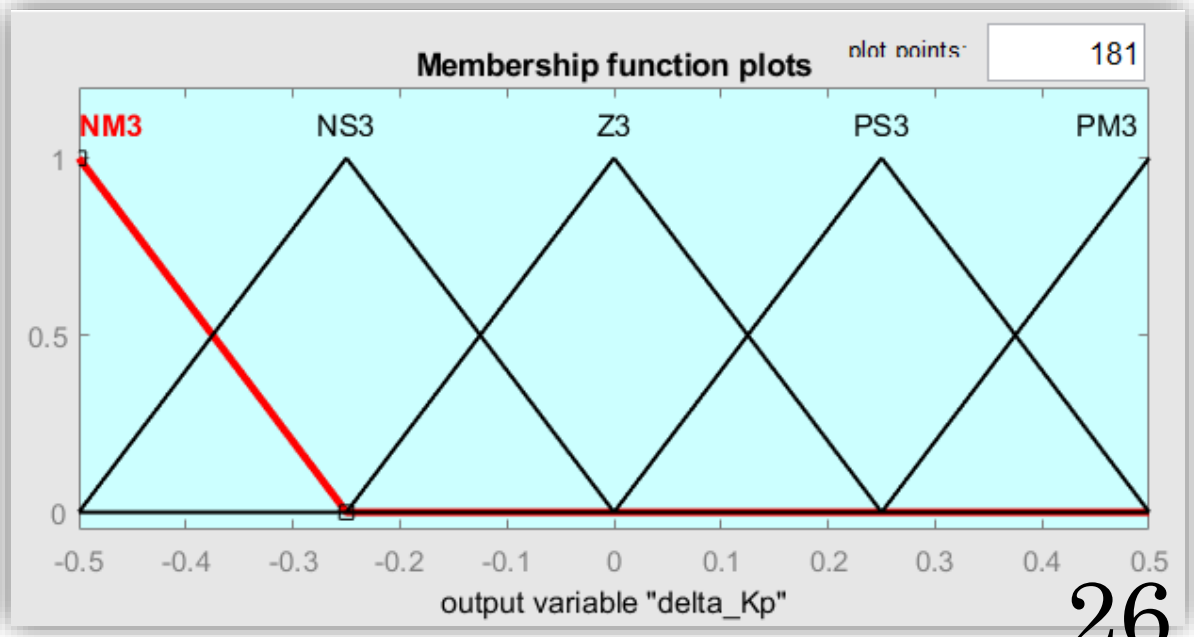


$$u(t) = (\Delta \mathrm{K}_p + -2)\,\mathrm{e(t)} + (\Delta \mathrm{K}_I + -0.01)\int e(t)\,\mathrm{dt} + (\Delta \mathrm{K}_D + 1)\frac{de}{dt}$$

Outputs


Membership function plots
plot points:
181
NM5
NS5
Z5
PS5
PM5
1
0.5
0
-0.5
-0.4
-0.3
-0.2
-0.1
0
0.1
0.2
0.3
0.4
0.5
output variable "delta_Kd"


Membership function plots
plot points:
181
NM4
NS4
Z4
PS4
PM4
1
0.5
0
-0.5
-0.4
-0.3
-0.2
-0.1
0
0.1
0.2
0.3
0.4
0.5
output variable "delta_Ki"


Membership function plots
plot points:
181
NM3
NS3
Z3
PS3
PM3
1
0.5
0
-0.5
-0.4
-0.3
-0.2
-0.1
0
0.1
0.2
0.3
0.4
0.5
output variable "delta_Kp"

Rule bases for Kp, Ki and Kd

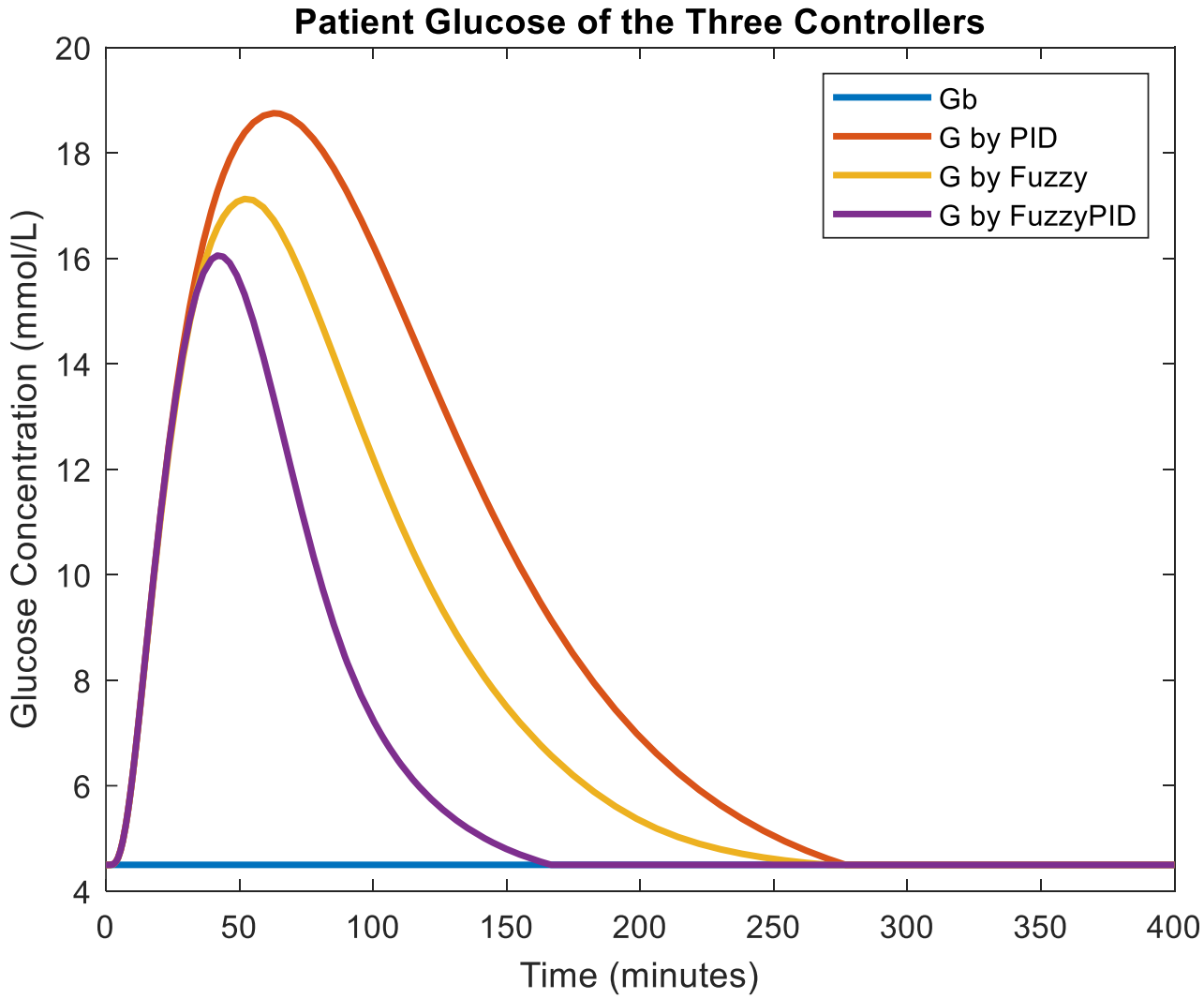
Delta_Kp	de					
e		NM2	NS2	Z2	PS2	PM2
	NL1	Z3	NM3	NS3	Z3	NS3
	NB1	NS3	NS3	PS3	Z3	NS3
	NM1	NM3	NS3	NS3	PS3	Z3
	NS1	NM3	NM3	PS3	NS3	PM3
	Z1	Z3	NS3	PM3	PS3	PS3

Delta_Ki	de					
e		NM2	NS2	Z2	PS2	PM2
	NL1	NS4	NS4	PM4	PS4	PS4
	NB1	PS4	PS4	NS4	Z4	PS4
	NM1	PM4	Z4	PS4	NS4	PM4
	NS1	PM4	PM4	PS4	NS4	NS4
	Z1	Z4	PM4	Z4	Z4	PS4

Delta_Kd	de					
e		NM2	NS2	Z2	PS2	PM2
	NL1	NM5	NM5	NM5	NM5	PM5
	NB1	NS5	NS5	NS5	PS5	PS5
	NM1	Z5	Z5	NS5	Z5	Z5
	NS1	PS5	PS5	NM5	NS5	NS5
	Z1	PM5	PM5	NS5	NM5	PM5

Results

Glucose responses of all the controllers



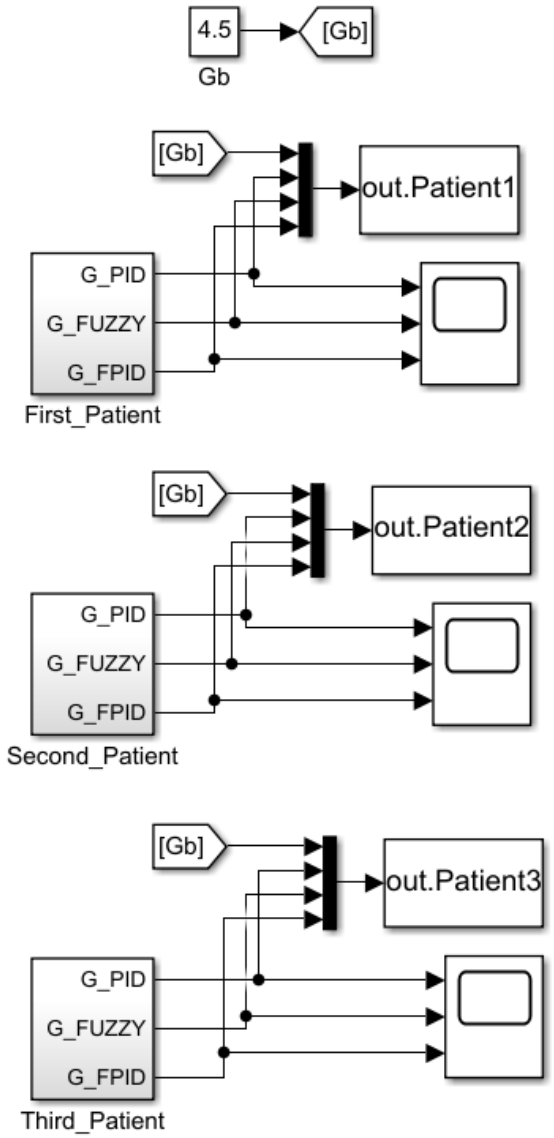
Performance Evaluation

Performance evaluation of the three controllers

Criteria	Type of Controller		
	PID Controller	Fuzzy Controller	FuzzyPID Controller
Peak (mmol/L)	18.75	17.13	16.05
Peak Time (min)	62.74	51.88	41.72
Settling Time (min)	276.86	270.12	166.67
IAE	1775.54	1260.54	746.63

Performance of the controllers under inter- and intra-patient variability via simulation

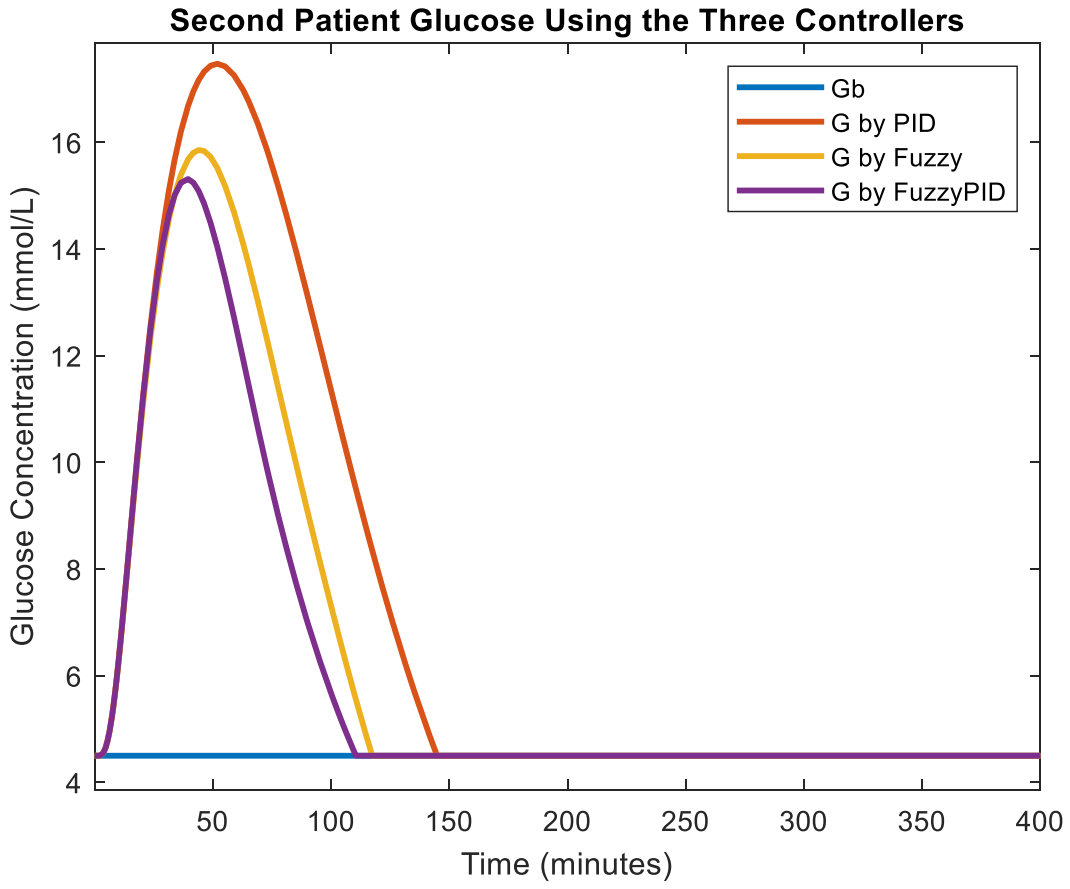
Block Diagram for three patients using PID, Fuzzy controller, and Fuzzy-PID

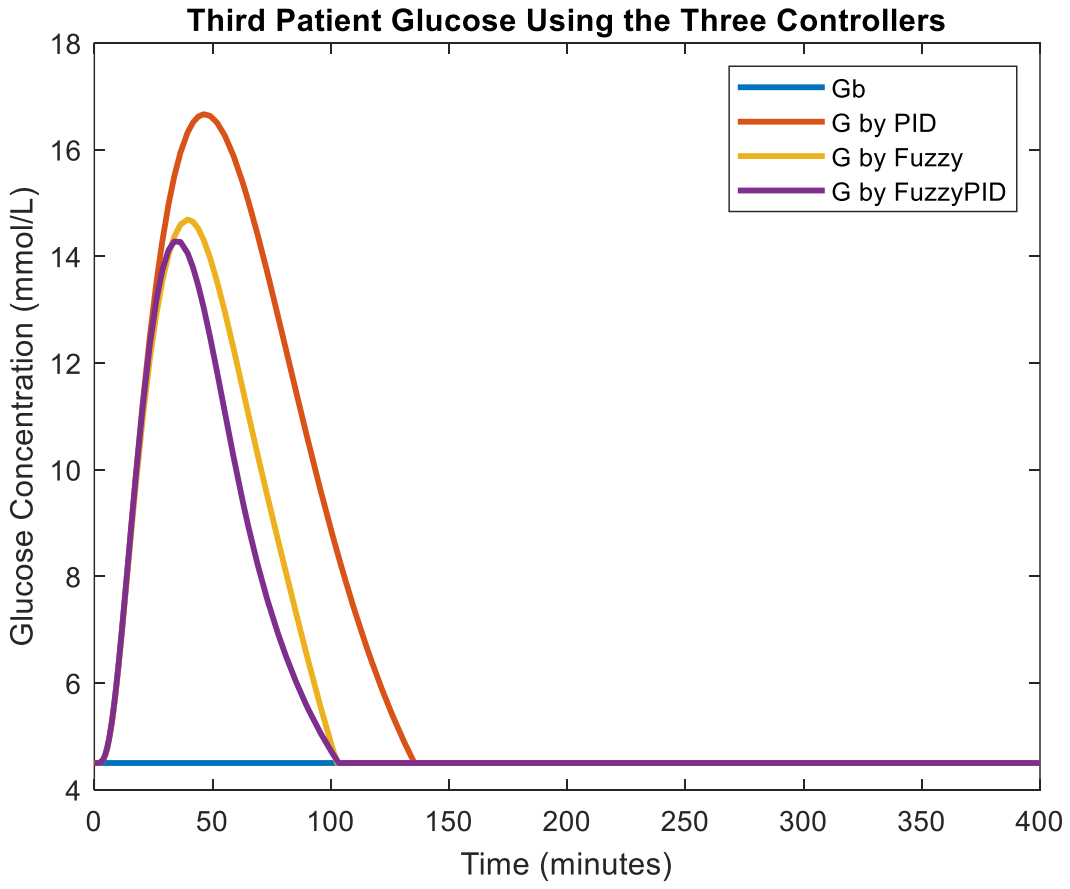


Patients' parameters

Parameter	Unit	Value		
		Patient 1	Patient 2	Patient 3
P_1	1/min	0	0	0
P_2	1/min	0.025	0.02	0.0072
P_3	-	$13x10^{-6}$	$5.3x10^{-6}$	$2.16x10^{-6}$

Results of the controllers under inter- and intra-patient variability via simulation





Conclusion & Future Work

Conclusion & Future Work

Since T1DM patients have no control over the blood glucose in their bodies because of their deficient pancreas. The Bergman Model was used for the simulation in MATLAB/SIMULINK To solve this problem, a Fuzzy-PID controller was designed, starting with the PID, then the Fuzzy controller, and finalizing with the Fuzzy-PID. The results showed that the PID controller can be used to control the blood glucose of a patient. The Fuzzy controller gave better responses. However, the Fuzzy-PID controller was the best of them all, as expected, and this was verified by the performance evaluation.

A possible improvement of the system could be achieved by the lookup table, as it will improve the performance of the system. Moreover, Lookup tables are better to be used in controllers because they can enhance the execution speed.

THANK YOU

Any questions ?